

Yi-Ting Lai

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EDUCATION

National Yang Ming Chiao Tung University

Hsinchu, ROC

MS Computer Science, *Advisor: Yi-Ping You* GPA: 3.92/4.3

Sep 2023 – Sep 2026 (expected)

- Thesis (tentative): **Compiler-Guided Exploitation of Block-Linear Memory Layout for Improving Intra-Warp Locality on GPUs**
- Courses: Advanced Compiler, Physical Design Automation, Accelerator Architectures for Machine Learning

National Taipei University of Technology

Taipei, ROC

BS Electronic Engineering GPA: 3.96/4.0, **Rank: 1/116**

Jun 2023

- **1st Place**, In-Campus Calculus Competition (245 participants, 2019)

EXPERIENCE

Advanced Semiconductor Engineering, Inc.

Kaohsiung, ROC

IT Intern

Jul – Aug 2022

- Automated defect detection using **YOLOv5**.
- Visualized factory production and downtime metrics by building data dashboards with **Apache Superset**.

TEACHING ASSISTANTSHIPS

NYCU Parallel Programming

Sep – Dec 2024, Feb – Jun 2025

NYCU Compiler Design, **Excellent Teaching Assistant Award**

Feb – Jun 2024

NYCU Introduction to Compiler Design

Sep 2023 – Jan 2024

NTUT Object-Oriented Programming

Feb – Jun 2021

PROJECTS

Compiler-Guided Exploitation of Block-Linear Memory Layout for Improving Intra-Warp Locality on GPUs | C++

Master Thesis

Implemented LLVM analyses and transformation passes that automatically map eligible GPU data to texture memory, exploiting the hardware-managed block-linear memory layout to improve intra-warp locality, thereby reducing cache-access wavefronts and alleviating memory-system contention.

VitaminC | C++

2023 – 2025

A C compiler targeting **QBE** and **LLVM IR**.

- Implemented key C features, including functions, arrays, pointers, loops, conditionals, structs, and unions.

Wav2Letter FPGA Accelerator | Verilog

2025

- Accelerated ASR inference by 15x, as measured by throughput, by designing a **systolic array** architecture.

sT Compiler | C

2023

A compiler for a self-designed language targeting **JASM** to run on the **JVM**.

- Designed a context-free grammar with documented syntax and semantics, implemented using **Flex** and **Bison**.

Concentration Judgment System | Python

2021 – 2022

A student concentration analysis system for online courses.

- Achieved real-time face and posture detection using **Dlib** and **OpenCV**.
- Implemented GUI and client communication with **PyQt5** and **WebSocket**.

SKILLS

Programming: C/C++, CUDA, Python, Bash

Software & Tools: LLVM, Git, NVIDIA Nsight Compute, Make/CMake, GDB, Docker, Linux

Language: Mandarin Chinese (Native), English (Professional Proficiency, TOEIC 875), Japanese (Basic)

PUBLICATIONS

1. Yi-Ting Lai, Meng-Shiuan Pan. “A Concentration Judgment System for Online Courses: Design and Implementation.” VTS Asia Pacific Wireless Communications Symposium (APWCS), 2023.